

L02 (OPTION B)

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ITAI-3377

1/28/26

## **Reflective Journal**

### **Challenges Faced**

One of the main challenges we faced during this lab was understanding the difference between standard TensorFlow models and TensorFlow Lite models. While TensorFlow is commonly used for training and testing models, TensorFlow Lite focuses more on deployment, which required us to shift how we thought about machine learning workflows. At first, it was not immediately clear why certain steps, such as model conversion and optimization, were necessary.

Another challenge involved understanding the optimization process. Although optimization improves performance and reduces model size, it was difficult to initially see how these changes affected the model internally. The conversion output messages and warnings required careful reading to ensure the process completed successfully. Interpreting these messages took time, especially since some outputs were technical and not always explained in detail.

We also had to ensure that the model continued to function properly after conversion. Verifying that inference still produced valid outputs was an important step, and it required attention to input and output tensor handling. Overall, these challenges encouraged us to be more careful, patient, and methodical throughout the process.

### **What We Learned**

Through this lab, we learned how machine learning models are prepared for real-world deployment using TensorFlow Lite. We gained hands-on experience converting a TensorFlow model into TensorFlow Lite format and applying optimizations to make the model more efficient. This helped us understand that building an AI model does not end after training; deployment considerations are just as important.

We also learned how to use the TensorFlow Lite Interpreter to load models and run inference. This reinforced our understanding of how AI models operate outside of traditional

development environments. Additionally, the lab helped us understand the importance of balancing model performance, size, and efficiency, especially when working with limited system resources.

Overall, this experience improved our understanding of the complete machine learning pipeline, from model development to deployment, and showed us how theoretical concepts are applied in practice.

### **Real-World Applications of TensorFlow Lite**

TensorFlow Lite plays a significant role in real-world AI applications, particularly on mobile and embedded devices. It is commonly used in mobile apps, smart cameras, wearable technology, and Internet of Things (IoT) devices. By running models directly on devices, TensorFlow Lite reduces the need for cloud-based processing, which helps lower latency and improve response time.

Running models on-device also improves privacy, since user data does not always need to be sent to external servers. This is especially important in applications such as facial recognition, voice assistants, and real-time image processing. The lab demonstrated how TensorFlow Lite enables efficient, fast, and practical AI solutions that can operate in real-world environments with limited hardware resources.

This experience helped us understand how TensorFlow Lite supports scalable and responsible AI deployments and why it is widely used in modern machine learning applications.